
Portfolio Component 3: Team Product/Prototype Proposal

Subject: SEDE Studios – Spring 2026

Course Coordinator: Valerie Gay, Dayna Sais, and Can Ding

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1. Introduction

Title of Product: Hybrid PV-Wind System with Grid Synchronization

Team name: Smart grid technologies

Team number: 1

Product Owner: Mohammad Abuhilaleh

1.1. Team Description

Name	Studio	Major
Hoang Gia Khiem Khuc	Application Studio B	Electrical and Electronic
Ba Huy Ta	Professional Studio B	Electronics Engineering
Duc Pham	Professional Studio B	Electrical and Electronic
Belal Abu-issa	Professional Studio B	Electrical Engineering
S M Redhwan Ahmed	Professional Studio A	Electrical Engineering
Aqib Mohamed Ameer	Professional Studio B	Electrical and Electronic

1.2. Project / Problem Overview

This project proposes the design and simulation of a grid-connected hybrid photovoltaic (PV) and wind system, focused on the control needed to synchronise its combined output with the utility grid. The current issue is that solar and wind are independently variable, so the inverter must hold synchronism and power balance as both inputs fluctuate, and must detect loss of grid supply before it energises a disconnected network and endangers line workers. Hence, we propose a DC-coupled design in which a PV array feeds a boost converter under Perturb and Observe MPPT and a permanent magnet synchronous generator (PMSG) wind subsystem feeds the same bus through a rectifier, with the shared DC link exported through a single three-phase Space Vector PWM (SVPWM) inverter using dq-frame PI current control and a Synchronous Reference Frame Phase-Locked Loop (SRF-PLL). Anti-islanding uses the Sandia Frequency Shift method, assessed by non-detection zone analysis against AS/NZS 4777.2:2020. Indicative ratings of 5 kW PV and 3 kW wind feeding a 400 V, 50 Hz connection through a 700 V DC link will be confirmed during initial sizing. The project is delivered entirely in simulation.

2. Product Requirements

2.1. Software, Information and Knowledge

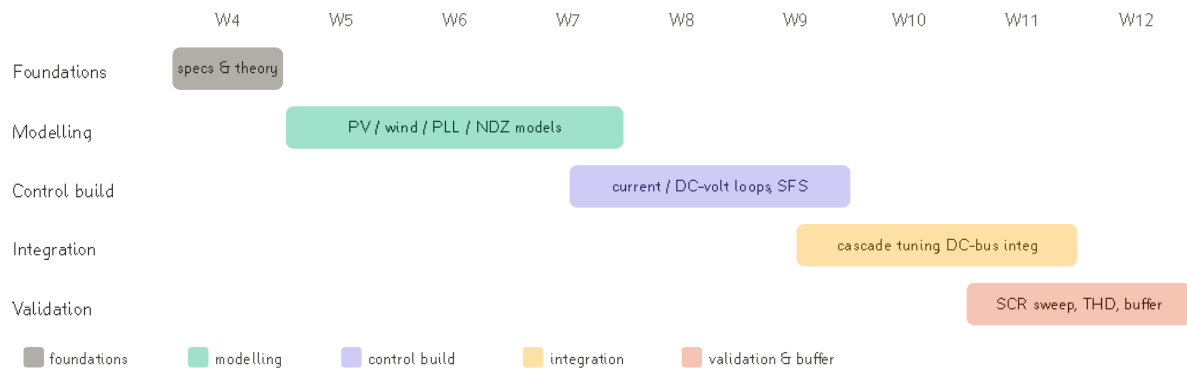
- **Required Skills:** Power electronics modelling, control design and PI tuning, dq/Park transformation theory, power system stability analysis, MATLAB/Simulink scripting, technical documentation.
- **Software:** MATLAB, Simulink, Simscape Electrical (Specialized Power Systems), GitHub.
- **Electronic components (modelled):** PV array, boost converter, PMSG, rectifier, DC-link capacitor, three-phase IGBT inverter (SVPWM), LCL filter, Thevenin grid equivalent.
- **Information and Knowledge:** AS/NZS 4777.2:2020, AEMO system strength reports, literature on PLL structures, dq current control, and islanding detection.

2.2. Success Criteria

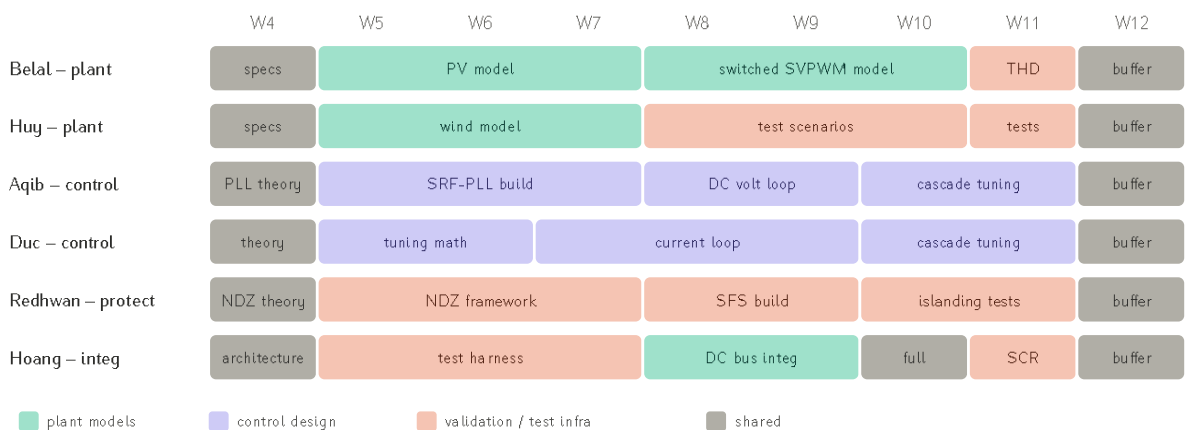
- Keep current THD below 5% at rated output.
- Limit DC-link voltage deviation to 5%, with recovery within 200 ms.
- Settle the inner current loop within 2 ms with under 10% overshoot.
- Re-lock the PLL within 100 ms after a 30° phase jump.
- Detect islanding and disconnect within 2 s.
- Maintain stable operation down to a short-circuit ratio of 3.

3. Development Timeline and Milestones

3.1. Project Timeline



3.2. Team Member Timeline



4. Path to Impact

As synchronous generation retires from the Australian grid, system strength is falling and inverters increasingly operate at connection points weaker than they were designed for, where the phase-locked loop and current controller can become unstable without any fault present. This project delivers a hybrid PV-wind system that meets AS/NZS 4777.2:2020 for power quality and anti-islanding, and characterises the short-circuit ratio at which its control loops lose stability, supporting the wider shift toward decentralised, low-carbon generation.

Expected challenges:

- **Controller tuning:** Maintaining stable PI and PLL performance across a wide range of irradiance and wind conditions, tuned in the correct inside-out sequence.
- **Anti-islanding sensitivity:** Balancing detection speed against nuisance tripping while meeting non-detection zone requirements.
- **Model fidelity:** Confirming that tuning derived in the averaged model holds once switching behaviour is introduced.
- **System integration:** Ensuring the PV, wind, and grid-interfacing subsystems operate as one coordinated control system.